BCS-084	Advanced Computer Network	L T P 3 0 0	3 Credits

Pre-requisites: students should have the basic knowledge of computer network.

Course Objectives:

CO1	To build an understanding of the fundamental concepts of computer networking
CO2	To introduce the basic taxonomy and terminology of computer networking.
CO3	To introduce advanced networking concepts.
CO4	To understand various Network Layer Protocol
CO5	To understand the concept of Socket Interface

Detailed Syllabus

MODULE-I

Introduction: Uses of computer Networks, Reference Models, Channel allocation problem, Multiple access Protocols, Ethernet, Wireless LANs, Broadband Networks, Structure Overlay Networks, P2P Computing

MODULE-II

Network Layer Design Issues, Addressing: Internet Address, Classful Addressing, Subnetting, Supernetting, Classless Addressing, dynamic Address Configuration, Network Layer Protocol: ARP, ICMP, IPV4 and IPV6.

MODULE-III

Transport Service, Elements of transport protocol, Process to Process Delivery, Internet Transport Protocols UDP, Internet Transport Protocols TCP, Performance Issues. The Application Layer: Server Model, Socket Interface: sockets, Connectionless interactive server, Connection-Oriented concurrent server.

Course Outcomes: After the completion of the course the student will be able to:

CO1	Enables the students to visualize the different aspects of networks, protocols and network design models.
CO2	Enables the students to examine various Data Link layer design issues and Data Link protocols.
CO3	Enables the students to analyze and compare different LAN protocols. Level 2,4
CO4	Enables the students to compare and select appropriate routing algorithms for a network
CO5	Enables the students to use Socket Interface

Text Books:

1. Computer Networks and Internets - Douglas E. Comer; PE.
2. Communication Networks - Leon-Garcia-Widjaja; TMH.
3. Internetworking with TCP / IP - Douglas E .Comer; PE.
4. TCP/IP protocol suite - ForouzanBehrouz A; TMH.
5. Computer Networks – Andrew S. Tannenbaum; PHI.
6. Data and Computer Communication - William Stallings; PHI